

MAE 263F – Homework 5

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I. PLATE MODEL SIMULATION

Question 1. Model the domain $l \times w$ as a thin plate of thickness h . Solve the static deformation until the configuration reaches steady state. Let the tip displacement at the centerline of the free edge $x = l$ be

$$\delta_{\text{plate}}(t) = z_{\text{tip}}(t) - z_{\text{tip}}(t = 0).$$

What is the steady-state displacement δ_{plate} ? Include a plot of δ_{plate} versus time t .

Answer: My simulation uses a time step of $\Delta t = 0.001$ s and a total simulation time of 20 s.

a) *Steady-state criterion:* Following the instructions, the system was considered to have reached steady state when the change in tip displacement over a one-second interval became smaller than 0.01%. Using this rule, the numerical simulation first satisfied the steady-state condition at

$$t^* = 8.053 \text{ s.}$$

b) *Steady-state displacement:* The corresponding steady value of the tip displacement is

$$\delta_{\text{plate}}^* = -0.03066 \text{ m.}$$

For comparison, the displacement at the final simulation time $t = 20$ s was

$$\delta_{\text{plate}}(20 \text{ s}) = -0.03520 \text{ m,}$$

which remains close to the steady value and continues the slow decay of oscillations.

c) *Tip displacement versus time:* Fig.1 shows the complete evolution of $\delta_{\text{plate}}(t)$ from $t = 0$ to $t = 20$ s. The point $t = t^*$ at which the plate first satisfies the steady-state condition is marked with a red circle, and a vertical dashed line indicates the moment at which steady state is detected.

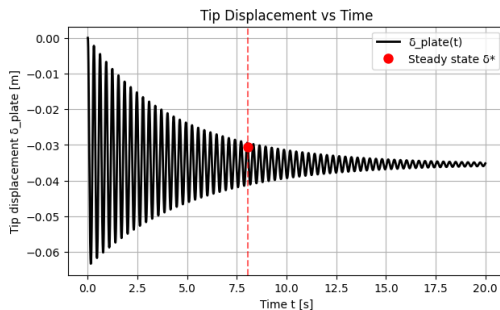


Fig. 1. Tip displacement $\delta_{\text{plate}}(t)$ versus time

I also include the plate configuration at the initial time $t = 0$ and at the final time $t = 20$ s, shown in Fig.2 and Fig.3.

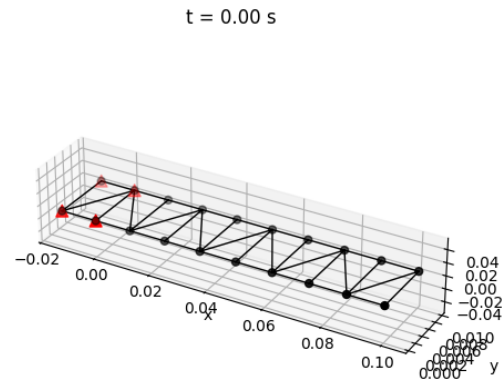


Fig. 2. Plate configuration at $t = 0$ s

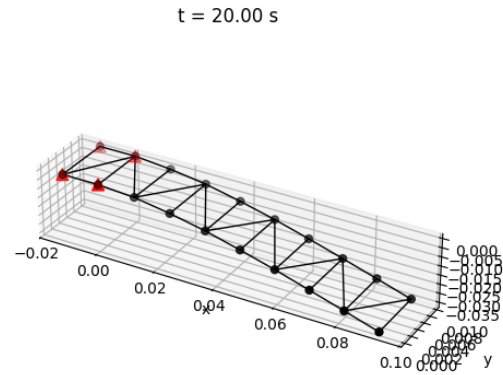


Fig. 3. Plate configuration at $t = 20$ s

II. COMPARISON WITH EULER-BERNOULLI BEAM THEORY

Question 2. For a cantilever beam under uniform load q (Newton per meter), the Euler-Bernoulli prediction for the

tip displacement is

$$\delta_{\text{EB}} = \frac{ql^4}{8YI}.$$

Compute δ_{EB} using the given material and geometric parameters. Compare the steady displacement from your discrete plate simulation (δ_{plate}) with the theoretical prediction (δ_{EB}). What is the normalized difference

$$\frac{||\delta_{\text{plate}}| - |\delta_{\text{EB}}||}{|\delta_{\text{EB}}|}?$$

Answer:

The Euler–Bernoulli prediction for the tip displacement is

$$\delta_{\text{EB}} = \frac{ql^4}{8YI} = 3.68 \times 10^{-2} \text{ m} \approx 0.03679 \text{ m}.$$

From the discrete plate simulation, the steady tip displacement was found to be

$$\delta_{\text{plate}} \approx -0.03520 \text{ m},$$

so that the magnitude is $|\delta_{\text{plate}}| \approx 0.03520 \text{ m}$.

The normalized difference between the plate model and the Euler–Bernoulli prediction is

$$\frac{||\delta_{\text{plate}}| - |\delta_{\text{EB}}||}{|\delta_{\text{EB}}|} = 4.33 \times 10^{-2} \approx 4.33\%.$$

Thus, the discrete plate result agrees with the Euler–Bernoulli beam theory to within about 4%.

CODE AND SOURCE ATTRIBUTION

The source code is adapted from *In class Lecture 14 - Main Discrete Shells Simple.ipynb* and has been refined and formatted with the assistance of ChatGPT.